Distilling a Language-Model Decision Policy into Browser-Scale Students

How a small model of structured triples can stand in for a three-billion-parameter decision model in a town-building game

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ABSTRACT

Hearthvale is a town simulation whose residents, households and institutions make repeated choices from structured questions. Its released decision system runs a compact set model in the browser for the kinds it knows and uses bounded hosted decisions for eligible questions it cannot answer locally or judges uncertain; local rules remain the final fallback. We distil a d1-3B reference teacher's answer distributions into a model of scope–attribute–value triples, with one token per fact and one per permitted answer. A shared town snapshot is cached across questions. In the evaluated core condition, the student reaches 0.0199 nats of teacher-to-student divergence on teacher-governed towns and about 70 microseconds per cached question on one processor core. Deep Sets also generalizes to the six withheld fact pairs tested, showing that preserving each fact as a unit matters more than attention alone. In 24 long simulated-player runs per policy, mean population and work-share differences were not detected, but the student's year-150 population spread was 1.54 times the teacher's; these tests do not establish equivalence. The released bundle covers `strike_resolve` in place of the experiment-only `where_to_settle` kind, which is now handled by the simulation. Its isolated extension scored 0.0149 nats on 397 held-out teacher-labelled strike questions and preserved exact outputs on 39,261 replayed questions from existing kinds. Long canary towns contained no strikes, so they do not measure its effect on strike outcomes. A daily training loop gathers selected questions, labels them with the teacher and publishes only candidates that pass model, browser and town-level checks. Hosted Anthropic narration is a separate, validated prose path and has no authority over simulation decisions.

1 Introduction

1.1 A town that decides for itself

Simulation games have long relied on hand-written rules to animate their populations. In Hearthvale the player governs a town as its council, while residents, households and factions make many smaller decisions. The released system serves most supported kinds from a student model in the browser. When a question is unfamiliar or the student judges its answer uncertain, an eligible session can consult a bounded hosted decision service using Liquid AI's `d1`; local rules cover unavailable answers. The game samples from each returned distribution, so small differences in

probabilities can accumulate over many years. The research evaluations below use a d1-3B reference teacher. Hosted `d1` is a distinct model, and the reported teacher-fidelity numbers do not measure agreement with that live service.

1.2 The cost of a hosted policy

The number of decisions rises with town size, game speed and concurrent players. A city asks about eighteen questions a second at normal speed and several hundred at the highest speeds. Serving every question remotely would consume shared inference capacity and add network delay. Fallback rules also change the simulation: in the long-run experiment here they produced markedly larger towns and far higher morning work share than the reference teacher. The browser student therefore carries the common path, with hosted requests bounded and local rules available when neither model supplies an answer.

The student learns the reference teacher's full probability distributions, rather than optimizing town outcomes directly. This separates policy imitation from game design: the objective is to retain the teacher's choice pattern while answering most questions without a network round trip. Teacher-labelled data from simulated and observed play feed the training loop, subject to validation and release gates.

1.3 Aim and contributions

Almost every decision input is categorical, such as whether food is low, whether rent is high or which institutions are active. The released bundle covers twenty-two decision kinds, including strike resolution. The central question is whether one browser-scale model can preserve the reference teacher's distributions on unseen towns and fact combinations, and whether repeated choices leave the course of whole towns sufficiently close to warrant use. We report the scope of each experiment separately because the core town evaluations preceded the strike extension.

We make four contributions. First, we adapt the Triple World Model's set-of-triples representation to typed decision questions. Second, we compare five model families and architecture ablations on teacher fidelity, cost and compositional generalization. Third, we evaluate repeated decisions through answer-level bias and complete-town outcomes, including their spread. Fourth, we describe a browser and Cloudflare training system that collects selected questions, obtains teacher labels and admits candidates through multiple gates. The strike extension provides a separate held-out test of adding a decision kind while preserving existing outputs.

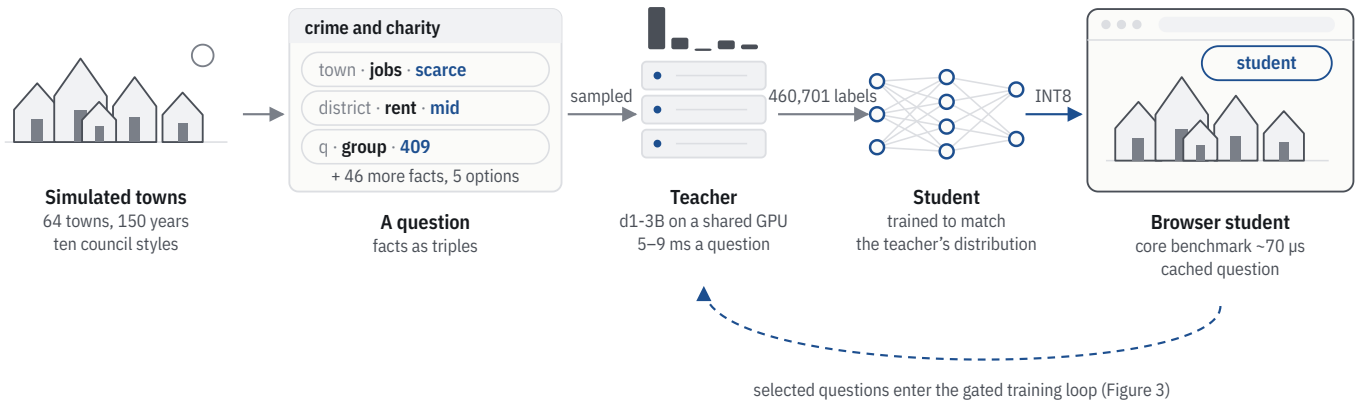


Figure 1. The core distillation experiment: structured questions from simulated towns are labelled by the d1-3B reference teacher, then used to train a browser student. The released system also has bounded hosted deferral and a separate daily training loop (Figure 3).

2 Background

2.1 Distillation

The idea of training a small model to imitate a large one is well established. Hinton and colleagues showed that a compact network trained on the full output distribution of a larger network, rather than on hard labels alone, retains much of its accuracy, because the distribution conveys how the teacher weighs the alternatives [1]. Rusu and colleagues carried the same idea to agents that act, distilling reinforcement-learning policies into smaller networks [2]. Our setting is a direct descendant of policy distillation, with one difference that shapes everything else: the teacher is a language model, and its inputs are structured records rather than images or game screens.

2.2 Learning on structured records

Structured records place the problem in the territory of tabular learning, where gradient-boosted decision trees such as XGBoost remain difficult to beat [3]. Neural approaches to tables commonly treat each column as a token and let a transformer relate them, as in FT-Transformer [4]. A known weakness of plain multilayer perceptrons is that they learn multiplicative interactions between features only with difficulty [5], which is why recommendation systems add explicit interaction terms. Factorization machines express every pairwise interaction as a product of learned feature vectors, and can therefore estimate the effect of a pair never observed together [6]. Set models, finally, treat their input as an unordered collection and are invariant to its order by construction [7].

2.3 Distribution shift and deferral

Two further strands bear on deployment. Imitation learning has long recognized that a copied policy visits states its teacher never showed it, and that its errors compound there; the DAgger algorithm addresses this by collecting states from the student's own behaviour and labelling them with the

teacher [8]. Selective prediction, in turn, allows a model to defer uncertain cases to a stronger one [9]. Both ideas reappear in the system, the first in how training data is gathered and the second in how the student consults its teacher.

2.4 The Triple World Model

The work reported here grows out of an earlier research system, the Triple World Model, which has not been published and is therefore described in some detail. Its premise is that a small model can reason well about a world if the world is presented to it as structure rather than as prose. A state is written as an unordered set of triples, each naming an entity, one of its attributes and the attribute's value, such as a pet, its hunger and the level of that hunger. A compact transformer reads all the triples at once and predicts the next state, producing every output triple simultaneously rather than one after another, and learning the change from the current state rather than the state afresh.

The system was built to test one claim: that this decomposition allows generalization to combinations of facts never seen together in training. On a benchmark spanning three domains, it predicted the outcome for novel combinations of entity and state with an F1 score of 0.74, and ablations showed that attention across positions was necessary to do so. Where the correct output depended on other triples present in the state, it reached an F1 of 0.978, and a model of 178,000 parameters matched one of 4.5 million. The same architecture, reduced to 29,000 parameters, ran a pet simulation entirely in the browser in 303 kilobytes of JavaScript with 98.9% exact agreement. The underlying argument is that structural priors, namely decomposition into triples, invariance to order and a shared embedding space, carry knowledge that an unstructured model would otherwise have to acquire through parameters and data.

2.5 Decision models

A recent class of models answers typed questions about a document with a probability distribution over permitted options in one pass. The d1-3B model supplies the reference labels used in this study; the game's bounded hosted path calls Liquid AI's `d1`, whose outputs are not evaluated here against those reference labels. Compared with general models that read free text, the student is specialized for Hearthvale's structured decision schema. This specialization makes a browser implementation practical but limits the model to kinds and vocabulary represented by its bundle.

3 Problem Setting

Each decision question consists of a kind, a shared snapshot and a short context. The kind names the decision, for instance a resident's routine, a household's choice to stay or leave, or a faction's stance on a proposal. The snapshot summarises the town, and for some kinds a district or a proposal, in coarse bands that change from one season to the next; many questions share the same snapshot. The context identifies who is asking: their trade, wealth, stage of life and immediate needs. The teacher's answer is a probability distribution over a fixed set of options for that kind, of which a question may permit a subset. Because the game samples from this distribution, a student must reproduce all of it rather than merely its most likely option, and we therefore measure agreement by the Kullback–Leibler divergence from teacher to student, expressed in nats.

A satisfactory student must meet three conditions. It must answer close to the teacher, including in towns it has never seen, since players build towns no training run anticipated. Towns under its control must behave like towns under the teacher, which is a stronger requirement than low average error, because the game samples from the student at every step for a century of game time and small biases accumulate. And it must be small and fast enough to run within the game's existing budget on an ordinary device, including the bursts of questions that open each game month.

4 Approach

4.1 Why the triple route

Hearthvale's decision questions are already structured records: a handful of categorical facts about a town, a district and the residents in question. The Triple World Model was designed for exactly this kind of input, and its properties correspond closely to the requirements set out above. Table 1 summarises the correspondence; the paragraphs below explain the most consequential points.

Table 1. Properties of the triple representation and the requirement each one serves.

Property	What it means here	Requirement it serves
Shared vocabulary	Every kind of question is written in the same terms, so one model answers all supported kinds	Size; maintenance
Order-free sets	Facts have no natural order, and lists of rules and institutions vary in length	Accuracy on varied towns
Multiplicative interactions	Attention relates pairs of facts, such as a faction and the interests a proposal serves	Accuracy
Compositional generalization	Answers remain sound for combinations of circumstances absent from training	Unseen towns
Small parameter count	The architecture has already run in a browser at tens of thousands of parameters	Device budget
Extension by tokens	A new fact is a new token and a new kind needs no feature code; the vocabulary can grow without retraining from scratch	Learning after release
A cacheable state	The snapshot, shared by many questions, can be encoded once and reused	Device budget; bursts

Two of these properties weigh most heavily. The first is compositional generalization. Players control the most consequential part of the input, the rules in force, and combine them in ways no simulated council ever did; a student that merely memorises combinations seen in training would fail precisely where the game depends on it. The Triple World Model's central result was generalization of this kind, and it depended on attention between facts rather than on a fixed function of their sum. The second is economy. A single representation for every kind of question, derived from the schemas the game already uses, means one small model rather than one per kind, one runtime in the browser and no feature engineering when a kind is added or changed.

The decision problem also differs from the world-modelling problem in ways that require adaptation. A world model predicts a whole next state, whereas a decision requires only a distribution over a few options, so the output is one score per option rather than a set of predicted

triples, and there is no dynamics step. And whereas a world model's state may hold several entities that share attributes, each scope in a decision question describes a single entity, which allows a triple to be represented as one token rather than three. We keep the representation and the principle of order-free sets, and change the output and the tokenization to fit the task.

4.2 Representation

Each question is rewritten as an unordered set of triples of the form scope, attribute and value, such as `(town, food, low)` or `(district, rent, high)`. A single function derives these triples from the schema that already defines each kind of question. Categorical fields become one triple each; integer fields with a small range are treated as categories; other numeric fields carry their value, compressed as sign(x)·ln(1+|x|), alongside the triple; and lists such as the active rules contribute one triple per item, together with a level where the item has one. The names of towns, districts and proposals are discarded, since nothing learned from a name transfers to another town. A value absent from the vocabulary maps to a shared unknown token, which also signals that the student is outside its experience. Because the representation follows from the schema, a new kind of question requires no new feature code.

4.3 The set model

The student is a set model with a single decision layer (Figure 2). Each triple is embedded as one vector, the sum of embeddings for its scope, attribute and value, with numeric values added along a learned direction for their attribute. The question's own triples, together with one token for each permitted answer, are the inputs of the decision layer. Each answer token is built from an embedding specific to that kind's option and one shared across kinds, so that an option such as leaving town draws on what is known about leaving in other decisions. In the decision layer, the question and answer tokens attend to one another and to the facts of the snapshot, which enter as their embeddings alone, normalised and projected into the keys and values that attention reads. Each answer token ends in one score, and a softmax over the permitted options gives the distribution.

Two properties of this layout make the model cheap to run. The snapshot's keys and values depend on nothing but the snapshot, so the browser computes them once and reuses them for every question that shares it, whatever its kind. And only the answer tokens reach the scores, so the layer computes their rows and no others.

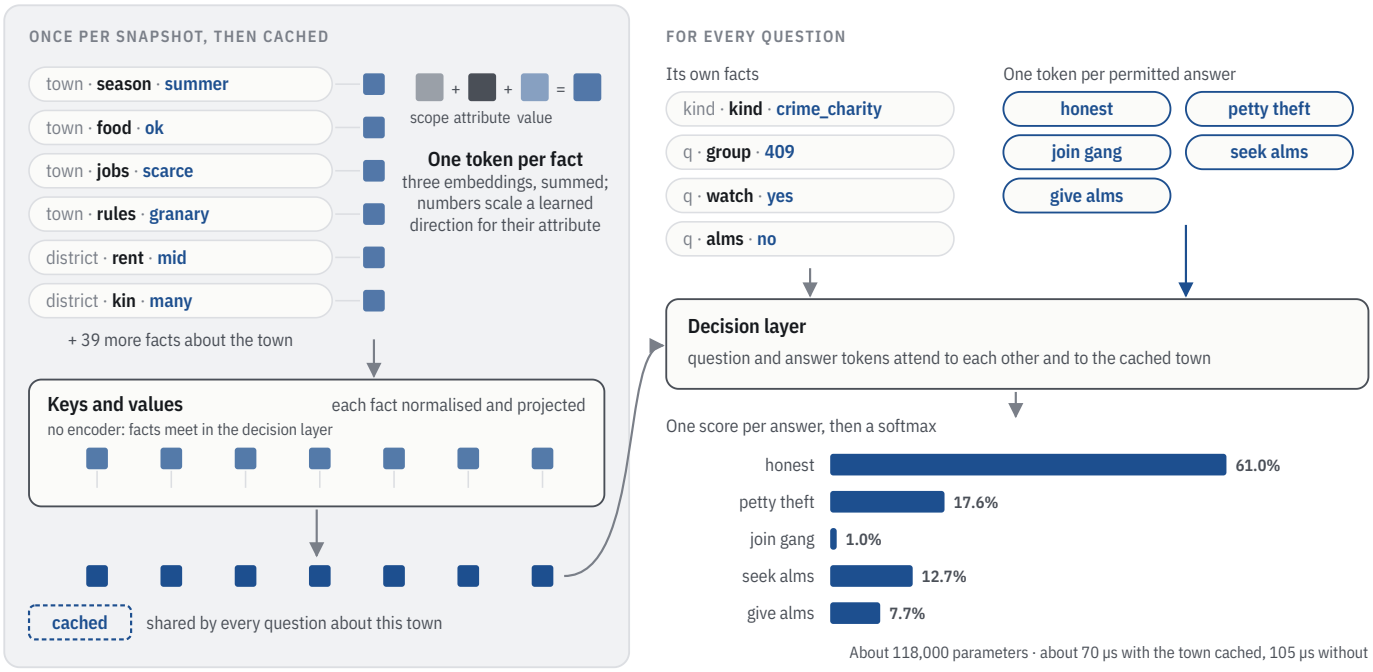


Figure 2. The set model, on one question: whether a group of 409 residents stays honest, turns to petty theft, joins a gang, seeks alms or gives them. Each fact becomes one token. The town's facts are projected once into keys and values and cached; each question runs only the decision layer, in which one token per permitted answer attends to the question's facts and to the cached town. The bars show the evaluated core student's answer to this question.

The snapshot has no encoder of its own. A variant that first relates the snapshot's facts to one another with an attention layer, as the Triple World Model does, is less accurate and three times slower when a town is first seen (Section 6.2): the decision layer, which already reads every fact from each answer token, relates them as each answer requires. Summing a triple's three embeddings into one token also departs from the Triple World Model, which gives each of entity, attribute and value its own token and binds them with position embeddings. In Hearthvale each scope describes one entity, so the sum already binds a value to its attribute and scope, and the single-token form needs a third of the tokens. We nevertheless retain the original layout as one of the alternatives, alongside three models representing the traditions described in Section 2: a Deep Sets model [7], in which a small network reads each triple's scope, attribute and value embeddings side by side, the triples' outputs are summed and a multilayer perceptron reads the sum, with dropout tuned for a fair comparison; a factorization machine whose option scores weight pairwise, and optionally third-order, interactions between triples, computed in linear time; and gradient-boosted trees, fitted separately for each kind on one-hot features. As an ablation we also keep a perceptron over the triple embeddings summed directly: since each embedding is itself the sum of its scope, attribute and value, that sum reduces a question to three bags of parts and loses which value belongs to which attribute.

5 Method

5.1 Data

Training data is gathered in two stages. The first provides breadth. Sixty-four towns are grown automatically to their 150th year across seven kinds of terrain, each governed by a stand-in council that rules on every proposal in one of ten political styles, from frugal to mercantile to pious. A recording layer captures every question exactly as the game client would pose it and answers it with the fallback rules so that the towns can proceed. Because these simulations do not step each resident individually, residents' immediate needs are static, and copies of each resident question with randomly drawn need levels let the student see the full range. These runs produce several million distinct questions spanning hamlets to late-era cities.

The second stage follows DAgger. Sixteen further towns are grown under student control, covering states induced by the student's own policy. The reference teacher labels sampled questions through the game's decision request path. A separate evaluation set comes from four teacher-governed towns. The core experiments use 460,701 teacher-labelled questions across the collection stages. Subsequent training also incorporates selected questions reported during play. The core experimental schema includes `where_to_settle`, which the simulation now handles itself. The released bundle instead covers twenty-one retained kinds plus `strike_resolve`. Thus the core experiment and the released bundle each cover twenty-two kinds, but their kind sets differ.

5.2 Training

Every neural student minimises the divergence from the teacher's distribution over the permitted options. Each question is weighted by the square root of how often the simulated towns asked it, so that common situations count for more without overwhelming rare ones. Training uses the AdamW optimiser with a one-cycle learning-rate schedule for twenty epochs, and keeps the epoch with the lowest error on held-out towns. The cheaper students can additionally learn from a trained set model: it labels 1.2 million further questions from the training towns, and the factorization machine and the Deep Sets model are trained on these labels together with the teacher's.

5.3 Evaluation protocol

Average error on held-out data is a necessary test but an insufficient one, so we evaluate the students in four increasingly demanding ways. The first measures error on towns excluded from training. Twenty per cent of towns are held out entirely, and because snapshots change every season, nearly every held-out question presents a snapshot never seen in training.

The second targets compositional generalization directly. For every pair of a question fact and a snapshot fact, we compare the teacher's average answer when both are present with an additive prediction from each fact alone, and choose four pairs whose joint effect departs most from additivity, together with two weakly interacting pairs as controls. Every training example containing both facts of a pair is removed, while examples containing either fact alone are kept, and we measure error on exactly the removed examples.

The third measures the student on questions from towns the teacher governs. These towns are grown headless, with residents' needs synthesised as in the first stage, so they are not play by people; but their states are those a town reaches under the teacher's policy, which is what a faithful student must handle, and they resemble the deployed game far more closely than towns run by the fallback rules. We call them teacher-governed towns. Alongside the divergence, weighted both uniformly and by how often each question is actually asked, we compare the average probability the student and the teacher assign to each answer, and the ratio of the two for answers the teacher gives rarely.

The fourth compares complete towns over 150 years. Runs share a map and initial state but receive different seeded random draws. A simulated player opens some dockets, lets others lapse and occasionally chooses at random; its random stream is kept separate from the residents' decisions. Four towns are grown six times each under the reference teacher, the evaluated core student, a training ablation without targeted young-town questions and fallback rules. For each outcome we report the mean shift, a bootstrap interval and a permutation p-value, as well as the ratio of within-town standard deviations. These experiments predate the strike extension and therefore establish neither its whole-town effect nor equivalence between the student and teacher.

5.4 Running in the browser

The browser loads a quantized student bundle with eight-bit weight matrices and a scale per row. Exported fixtures check parity between the training implementation and JavaScript before publication. The simulation worker caches each snapshot's projected keys and values, as well as repeated answers; the game samples after retrieving a distribution, so answer caching does not fix the random outcome. A per-turn work budget spreads monthly question bursts over subsequent steps. A play session pins the published bundle it loaded, while later sessions can take a newer manifest entry. Unknown kinds use bounded hosted decisions when eligible and local rules otherwise.

Hosted Anthropic narration has a separate role: it turns bounded simulation facts into prose for Council cards and inspector notes. Opening Council queues its surfaced cards, with at most two requests in flight; saved prose is reused. The card path checks completed sentences for required request details and decision stakes before streaming a preview; an incomplete or invalid response falls back to template prose. Narration cannot choose an option, alter the probabilities or mutate the simulation policy.

5.5 Learning after release

A fixed student would eventually meet towns its training never covered, so the system keeps learning from labelled questions. A small auxiliary network, trained on held-out towns, predicts the student's own error from the answer tokens' final states, answer entropy and the share of unknown inputs. This estimate guides reporting and live referral. The released Cloudflare training workflow requests hosted `d1` labels; the core experiments below were labelled by the d1-3B reference teacher, so their fidelity estimates are specific to that reference policy.

The browser reports questions with high predicted error, unseen values or unsupported kinds. Reports contain validated game-state questions and are deduplicated; the intake limits contributions from one client. A separate paid path can refer eligible uncertain questions to hosted `d1` under individual and shared quotas. Those live `d1` answers are not interchangeable with the d1-3B reference labels used for the core fidelity experiments.

A Cloudflare-scheduled coordinator runs the training workflow daily when enough eligible questions have accumulated. The container obtains hosted `d1` labels, trains a candidate and submits it to the publication gates. The loop admits one valid report of a question, while limiting per-client intake and unusually composed states. It does not change the model within an open play session.

What one player can contribute is bounded instead. A cycle takes at most 400 questions from any one client, in the order they arrived, and defers the rest to later cycles; questions whose categorical facts form more than two pairs never produced by any simulated town may make up at most a quarter of a cycle unless three clients reported them; and questions already trained on are skipped. A cycle can also take rows gathered on purpose, such as teacher-labelled questions from young towns the student governs.

For ordinary updates, training begins from the published weights and extends vocabulary entries as needed. Architecture changes require training on the accumulated data. Gates compare teacher-governed error overall and by traffic, frequent-kind error, answer bias including young towns, rare-answer ratios, held-out new questions, JavaScript parity, bundle size, speed and simulated-player canary towns. A candidate is published only when every required check passes. The isolated strike extension froze the shared model and confidence head and trained 288 strike-specific embedding parameters; its separate results appear in Section 6.9.

During eligible play, the browser can refer a limited share of its least certain questions to the hosted decision service. A per-player rate limit and shared server budget bound this path. If the request is refused or unavailable, the already computed student distribution remains usable. This deferral changes some live answers but does not turn hosted `d1` into a validated measurement of the d1-3B reference teacher.

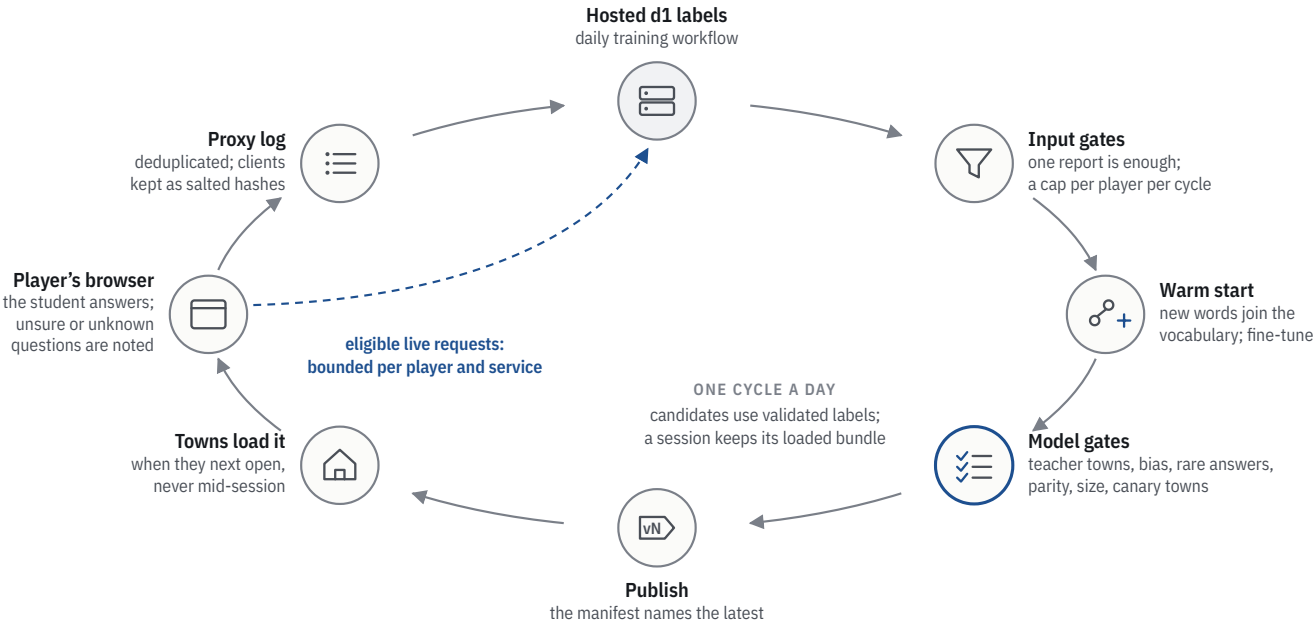


Figure 3. Daily training and publication. Validated reports and teacher labels feed a candidate; model, browser and town gates control publication. A play session keeps its loaded student bundle.

6 Results

6.1 Architectures compared

The five families are compared on the broad first-stage data under identical conditions (Table 2, Figure 4). The set model reproduces the teacher most closely, with a divergence of 0.025 nats on towns it never saw and agreement on the most likely answer in 86% of questions. Gradient-boosted trees come close in accuracy, but a model covering every kind requires about 12.7 megabytes of trees, some forty times the size of the set model. The factorization machine is somewhat less accurate yet exceptionally cheap, and learning from the set model's labels closes part of the gap. The Deep Sets model comes closest of the alternatives: its best configuration reaches 0.0270 nats, 5% above the set model with one layer per stage and 7% above the one with two, with about three times the parameters; trained on the set model's labels as well, it reaches 0.0256, with about four times the parameters. Its smallest configuration answers a cached question in under half the set model's time, at 14% more divergence. The pooled ablation, which sums every fact's parts before any nonlinearity, is the least accurate by a wide margin. Within the set model, a single layer in each stage costs almost nothing in accuracy against two and takes about a third of the time.

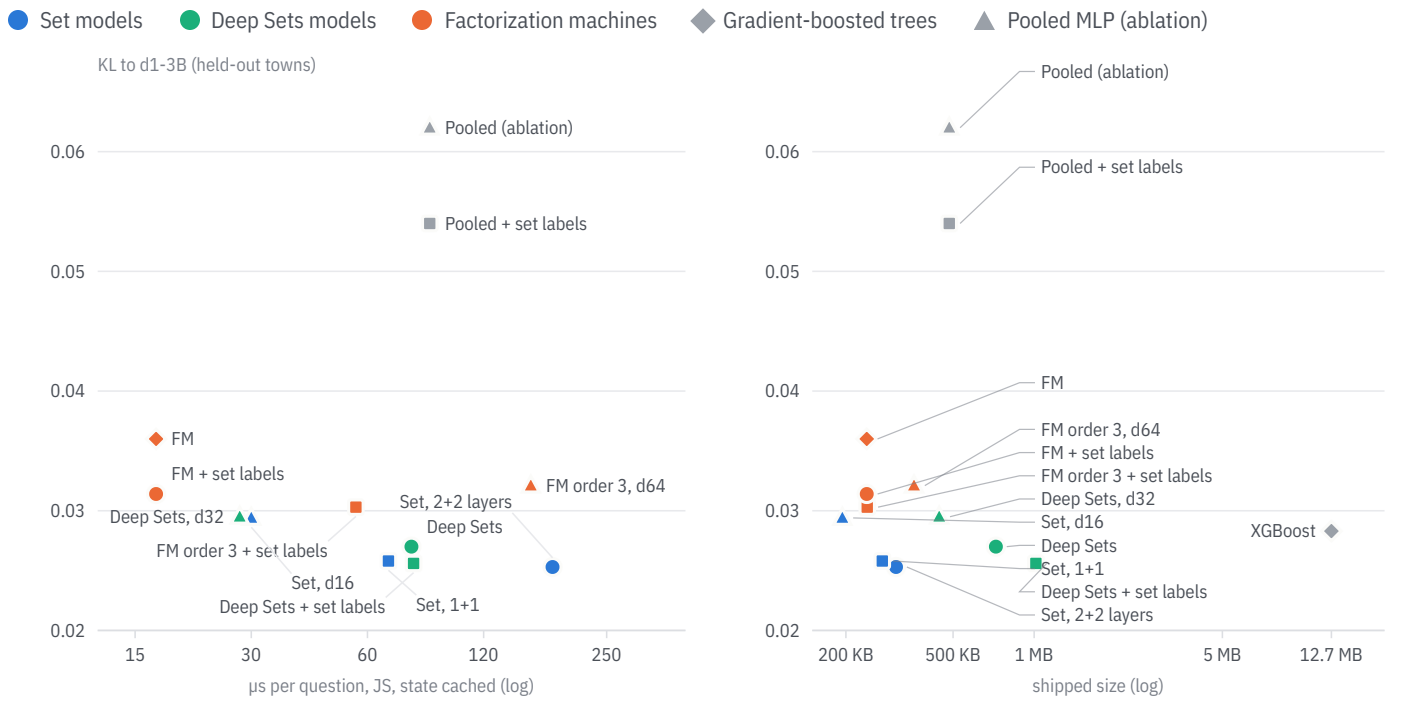


Figure 4. Divergence from the teacher on held-out towns of the first stage against cost per question in the browser with the town cached (left) and the size of the bundle (right). Lower and further left is better. Each model keeps its colour and marker in both panels. Gradient-boosted trees have no browser implementation and appear only on the right.

Table 2. Architectures compared on 30,667 questions from held-out towns of the first stage. The baseline, which answers every question with its kind's average distribution, scores 0.150. Parameters: set models 49,000 to 129,000; Deep Sets 128,000 (small), 338,000 (best) and 471,000 (with set-model labels). The pooled MLP is an ablation that sums every fact's parts before any nonlinearity.

Student	Divergence (nats)	Top answer agrees	Bundle size	Time per question, town cached
Set model, two layers per stage	0.0253	85.7%	307 KB	~180 µs
Set model, one layer per stage	**0.0258**	**85.3%**	**273 KB**	**~70 µs**
Set model, small (d = 16)	0.0294	84.4%	194 KB	~30 µs
Deep Sets, with set-model labels	0.0256	85.8%	1.0 MB	~80 µs
Deep Sets, best of four (d = 64)	0.0270	84.7%	721 KB	~80 µs
Deep Sets, small (d = 32)	0.0295	84.2%	444 KB	~30 µs
Gradient-boosted trees	0.0283	85.4%	~12.7 MB	not built
Factorization machine, with set-model labels	0.0314	84.0%	239 KB	~17 µs
Factorization machine	0.0360	82.6%	239 KB	~17 µs
Pooled MLP, best of four (ablation)	0.0620	80.5%	484 KB	~87 µs

6.2 Which parts of the set model earn their cost

Most set-model parameters belong to its vocabulary. Table 4 compares architecture conditions trained with the same recipe on the core data, changing one component at a time from a set model with a snapshot encoder. Timings use the same 2,000 questions from teacher-governed towns. These are architecture measurements, not a benchmark of the released strike extension.

Removing the snapshot encoder improves each measured core-data metric. Relative to the encoder condition, held-out error falls 5%, teacher-governed error 4% and traffic-weighted error 9%; uncached questions take about a third as long. The result supports the released architecture's single decision layer, though it does not attribute the gain to one mechanism with certainty.

Narrowing the model trades accuracy for speed roughly in proportion. At a width of 24 the error rises by about 5% on held-out towns and 8% on teacher-governed towns, and cached answers come a third faster; at 20 the error rises by 8% to 12% for answers nearly twice as fast. A single attention head saves almost nothing and costs a little accuracy. Training the narrower model on the set model's labels as well recovers most of its loss, but those labels bring a larger vocabulary with them, and the bundle grows by half.

Table 4. Variants of the set model, full vocabulary, the same recipe and data. Parameters exclude the error estimate (2,209); the evaluated core student has 119,201, because it was trained in the learning loop on these data and on questions from play, which add 83 vocabulary entries (2,656 parameters). Held-out towns: the training split; teacher-governed towns: 40,119 questions, overall and weighted by traffic, with the largest bias on any answer of a frequent kind in points. Times per question on 2,000 teacher-governed questions on one core, with the town cached and without, before the per-fact cache of Section 7.2; download is the gzipped bundle. ff: feed-forward width.

Student	Parameters	Held-out towns	Teacher-governed	By traffic	Largest bias	Cached	Uncached	Download
With a snapshot encoder (width 32, ff 64)	125,089	0.0217	0.0204	0.0122	2.1	67 µs	314 µs	172 KB
No snapshot encoder (core condition)	**116,545**	**0.0206**	**0.0196**	**0.0111**	**2.0**	**67 µs**	**105 µs**	**162 KB**
No encoder, ff 32	114,465	0.0207	0.0199	0.0125	3.8	60 µs	97 µs	160 KB
ff 32	120,929	0.0219	0.0211	0.0115	2.0	60 µs	269 µs	168 KB
Width 24, ff 48	89,977	0.0229	0.0220	0.0128	1.7	45 µs	203 µs	135 KB
Width 24, ff 48, one head	89,977	0.0233	0.0230	0.0132	1.8	45 µs	194 µs	135 KB
Width 20, ff 40	73,381	0.0235	0.0229	0.0128	2.4	36 µs	155 µs	118 KB
Width 24, ff 48, set-model labels	136,081	0.0210	0.0209	0.0122	1.5	46 µs	201 µs	200 KB

6.3 Compositional generalization

The test of withheld pairs separates the ways of combining facts more than the architectures (Table A1). On combinations of facts it never saw together, the set model's error is 4% higher than on ordinary questions (0.0248 against 0.0238 nats), and the Deep Sets model's is no higher at all (0.0268 against 0.0271); in absolute terms the set model stays slightly ahead on the withheld rows, as on ordinary ones. The factorization machines lose between 13% and 15%, consistent with their design,

and they perform best of all on the purest two-way interactions, such as craftsmen weighing a proposal that favours craft. The pooled ablation's error, by contrast, nearly quadruples, and on one pair, young people choosing a trade when farms offer no work and wellbeing is poor, its answers are scarcely better than the kind's average: having summed every fact's parts first, it cannot tell farms without work from other combinations of the same words. On this test, which covers six pairs, attention is not required; reading each fact whole is.

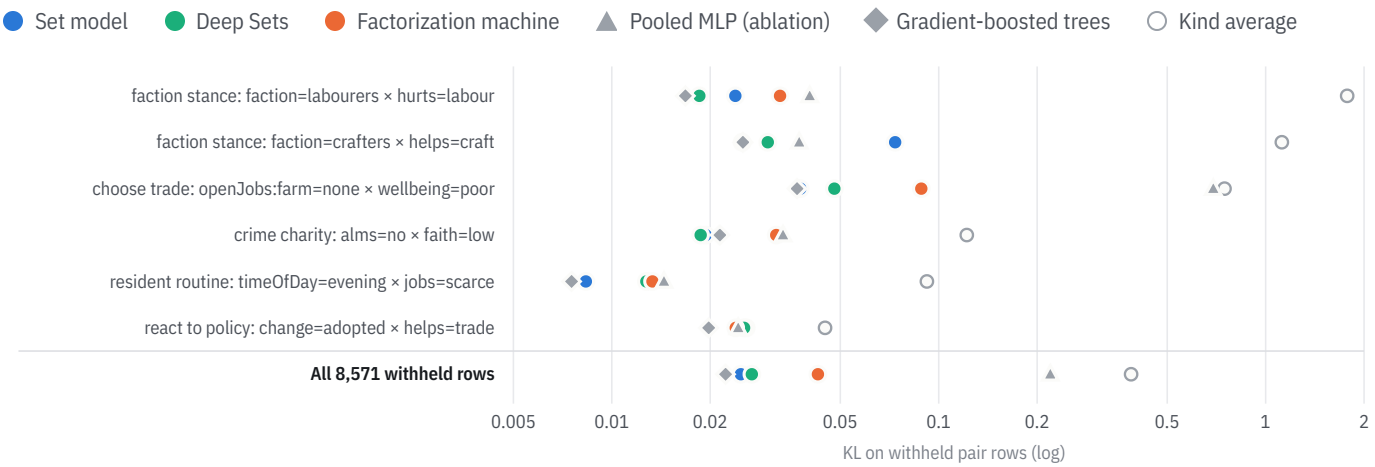


Figure 5. Divergence on questions whose pair of facts was withheld from training (logarithmic scale). Hollow markers show the kind-average baseline.

6.4 The evaluated core student

Table 3 reports the evaluated core student and alternatives on held-out and teacher-governed towns. The core student reaches 0.0199 nats on teacher-governed questions and 0.0109 when weighted by question traffic. For daily routine questions, its mean answer probabilities differ from the reference teacher's by at most 0.6 percentage points (Figure 6). These evaluations measure the core decision kinds and the d1-3B reference teacher; they do not measure strike resolution or the live hosted `d1` path. Removing on-policy training raises teacher-governed error, particularly on household decisions (Table A2), supporting collection from student-reached states.

Table 3. The evaluated core student and the main alternatives, trained on both stages of data. Teacher-governed towns: 40,119 questions from four headless towns the teacher governed, never used in training. The second row is the same student trained without the young-town questions of Section 6.6.

Student	Held-out towns	Top answer agrees	Teacher-governed towns	Same, by traffic
Set model, no snapshot encoder (core)	**0.0203**	**87.2%**	**0.0199**	**0.0109**
The same, without young-town questions	0.0206	86.9%	0.0203	0.0114
Set model with a snapshot encoder	0.0217	86.4%	0.0204	0.0122
Set model, small (width 16)	0.0256	84.7%	0.0265	0.0155
Factorization machine, with set-model labels	0.0256	85.4%	0.0265	0.0156

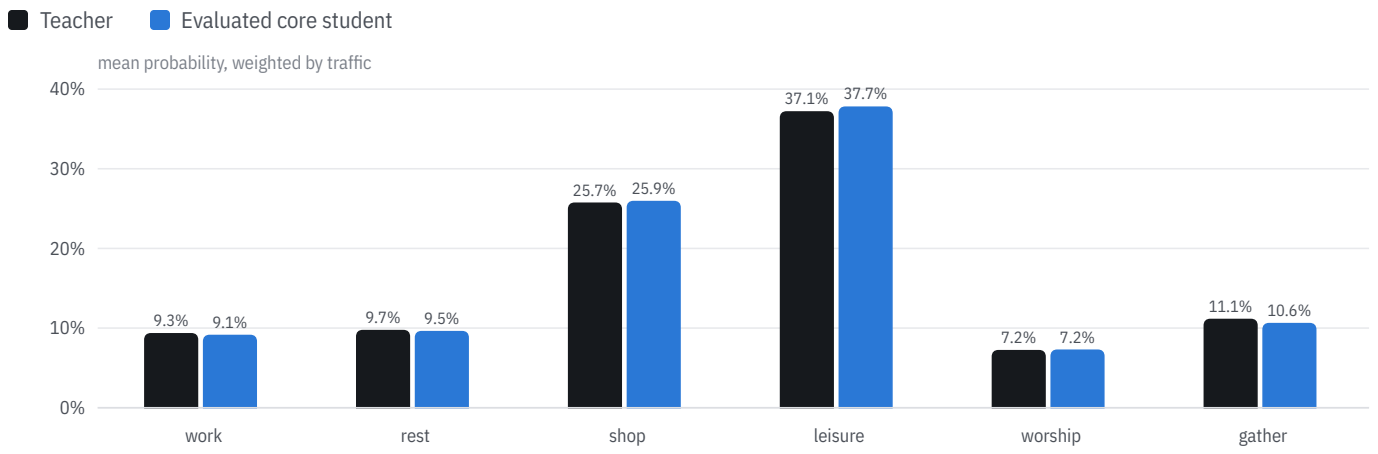


Figure 6. Traffic-weighted mean probability of each daily-routine answer on teacher-governed questions: the reference teacher and evaluated core student.

6.5 Behaviour of complete towns

The long-run core-policy test asks whether towns behave alike. Chance alone shapes a town's history strongly: grown six times under the teacher, one town ends its 150th year with anywhere from 561 to 2,996 inhabitants. Table A3 and Figure 7 compare runs of the same four towns, played by the simulated player, under the student and under the teacher.

Across these 24 runs per policy, measured mean shifts are +0.08, +0.28 and +0.07 in log population at years 50, 100 and 150 (permutation p-values 0.68, 0.17 and 0.71). Morning and evening work-share shifts are +0.3 and less than 0.1 percentage points ($p = 0.63$ and 0.98), and year-150 wellbeing shifts by +2.5 points ($p = 0.32$). The tests did not detect those mean differences, but their intervals allow meaningful effects; they are not equivalence tests. Fallback rules yield much larger towns and a 36-point higher morning work share.

In spread, the student's towns mostly vary as much as the teacher's: the ratio of within-town standard deviations lies between 0.75 and 1.08 for population at the 50th and 100th years, both work shares and wellbeing, with intervals that include one. The exception is population at the 150th year, which varies half again as much (1.54, interval 1.12 to 2.30): in two of the 24 runs a town grows to more than 9,000 inhabitants where none of the teacher's runs of it passes 3,000. Individual answers do not explain this; across the teacher-governed questions, for instance, the student's probability of work in the morning varies 0.96 times as much as the teacher's. Something in the student's play compounds over a century in some towns, and finding it is the next question for the evaluation of Section 7.4.

A small architecture comparison grows each town twice with a set model and a Deep Sets model trained from scratch on the same data. Their mean shifts were not resolved at this sample size, while the Deep Sets runs ranged from 56 to 8,536 residents at year 150. The larger core-student experiment and this two-run comparison answer different questions; neither estimates the whole-town effect of strike resolution.

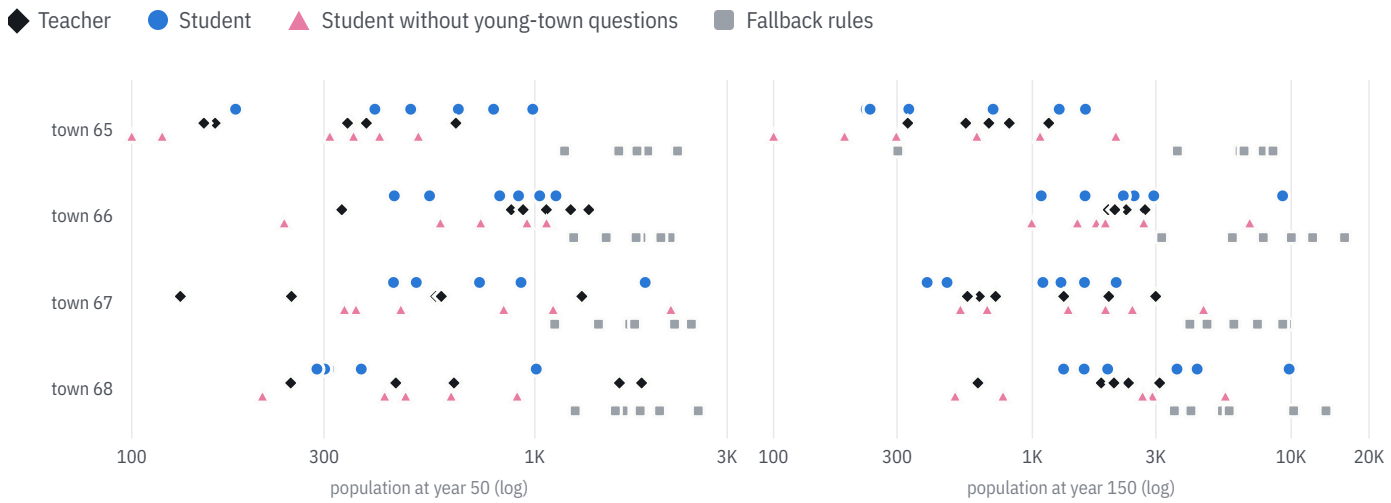


Figure 7. Population at years 50 and 150 in core-policy experiments on four towns, six runs per policy. The no-young-town condition is a training ablation. These runs do not include the released strike kind.

6.6 Young towns

Young towns are a small part of what a student learns from, about one question in six of the teacher-governed set, because a town does not stay small for long. Trained without questions gathered there on purpose, the student leans. Comparing each answer's average probability under that student and the teacher on questions from the hamlets and villages of the teacher-governed towns, one decision stands out: a household's choice whether to stay. There it gives staying 64.5% against the teacher's 69.5%, and emigrating 10.9% against 9.3%; no other decision bearing on growth, among arrivals, homes, land, enterprises and trades, differs by more than about two points. In older towns the same decision shows no lean (64.3% against 63.8%).

The targeted young-town ablation probes whether small answer biases compound. In one simulation condition without foundation costs, the student lacking young-town questions produced towns about a third smaller than the teacher's at year 50 (−0.43 in log population, 90% interval −0.65 to −0.22, $p = 0.004$), a lag recovered by year 100. Under the core town-evaluation condition, the year-50 shift was −0.11 ($p = 0.56$). The interaction between policy bias and simulation conditions is research-relevant: a small answer bias can matter greatly or be absorbed by other mechanics. Neither condition evaluates strike resolution.

Targeted training adds 5,840 teacher-labelled household questions from student-governed young towns and gates the largest answer bias in hamlets and villages. In the evaluated core condition, the staying bias in young towns falls to 2.4 points; across all towns the student assigns 65.1% to staying against the teacher's 65.1%, and 9.6% to emigration against 9.3%.

The same comparison shows where answers are too even. In the evening the student's probability of work moves only 0.40 times as far as the teacher's from one set of circumstances to another. At 3 to 4 percent of evening answers the compression is too small to change towns, but it is the kind of quiet distortion that a standing comparison of answers by kind, town stage and responsiveness should watch as both the game and its students change.

6.7 Cost

In the measured core configuration, a cached question takes about 70 microseconds on one processor core, around 200 microseconds with nothing cached. The then-measured bundle occupied about 356 kilobytes, 167 kilobytes compressed. These figures describe the benchmarked core bundle; the released strike-capable bundle is 364,786 bytes and its separate parity benchmark measured 91 microseconds with the state cached on its test machine. The training and runtime environments differ, so those timings should not be read as a controlled speed comparison.

6.8 Confidence and the learning loop

On teacher-governed core questions, the error estimate has rank correlation 0.53 with actual divergence. Marking the 5% it judges most uncertain captures 46% of the worst-error questions, nine times random selection (Table A4). This supports selective reporting and bounded deferral, while also showing that more than half of the worst questions escape that cutoff. The daily loop uses such signals and evaluates each candidate before publication; its gates are a safeguard, not proof of behavioral equivalence.

6.9 Strike resolution in the released bundle

The released twenty-two-kind bundle covers `strike_resolve` and twenty-one retained kinds. The core experiments elsewhere in this paper also include `where_to_settle`, a kind now handled by the simulation, so they are not a direct evaluation of this bundle. For this extension, 288 strike-specific embedding parameters were trained while the shared weights and confidence head were frozen. On 397 held-out teacher-labelled strike questions, divergence is 0.0149 nats and top-answer agreement is 93.5%, against 0.0514 nats for the kind-average baseline. A replay of 39,261 questions across the existing kinds found identical probabilities and expected-error estimates to the frozen core bundle. The bundle passed JavaScript parity, size, speed and short-town publication gates. Twelve 40-year canary towns asked no strike questions, so their unchanged outcomes show preservation of existing behavior only. Strike coverage drew on 72 observed snapshots from three map seeds, with schema-valid context variations; calibration of the frozen confidence head for strikes remains untested.

7 Discussion

7.1 What the triple route contributes

Most of the teacher's answers depend on individual facts and on pairs of facts, such as a faction and the interests a proposal favours, or a trade and the jobs available. The first requirement this places on a student is that each fact reach it whole. The set model, the factorization machines and the Deep Sets model all keep a fact's scope, attribute and value together before facts are combined, and all three handle combinations they never saw; the pooled ablation, which sums every fact's parts first, cannot tell which value belongs to which attribute and fails on exactly those combinations. Among models that read facts whole, the differences are of degree. Factorization machines capture pairs efficiently but give each pair a fixed strength, and lose most on the withheld pairs. Attention can make an interaction depend on a third fact and disregard irrelevant ones, but a Deep Sets model,

pooling nonlinear encodings of each fact, generalizes to the pairs tested as well. In the configurations tested, the set model's advantage is efficiency and a modest margin of accuracy: the lowest divergence, with about a third of the Deep Sets model's parameters and a quarter of its cost with nothing cached. Separating the architectures further would take a wider test, with more pairs and interactions of three facts, and several training seeds.

The representation is economical in practice. One set model serves the supported decision kinds; shared snapshot projections keep repeated questions cheap; and vocabulary rows can be added without changing the question architecture. The released strike extension demonstrates a narrower kind-addition route: freeze shared parameters and fit only kind-specific rows while verifying exact parity on existing kinds.

7.2 Why the student is fast

The evaluated core student answers in well under a tenth of a millisecond on one processor core because of four architectural choices. Its roughly 119,000 parameters describe the measured core bundle. The released strike extension trained 288 kind-specific parameters while holding the rest of its own base bundle fixed.

The first is that the input is never text. The teacher reads some three hundred tokens of prose for each question and must interpret them with three billion parameters before it can weigh the options. The student receives the same information as about fifty facts, each already a single token, so reading a question reduces to looking up three vectors per fact and adding them. This is also why most of the model's parameters cost nothing to run: the table of value embeddings holds nearly three quarters of them, and a question touches only the rows its facts name (Table 5).

The second is that the town is read once. A town's facts change once a season, while dozens of questions of every kind are asked about it each month. The keys and values the decision layer reads from them are computed when a snapshot is first seen and reused by every question until the season turns, so a question pays only for its own few facts and its answers. Because each fact's keys and values depend on that fact alone, they are kept as well, so a new town made of facts met before, as most are, costs about what a cached one does.

The third is that permitted answers are tokens. Each enters the decision layer with a kind-specific and a shared embedding and leaves as one score. The runtime computes only the answer rows needed for the question. This allows a single model to serve supported kinds without a separate output network per kind.

The fourth is that the model is small enough to need no special machinery. With a width of 32 and one layer, the arithmetic per question suits plain JavaScript loops over typed arrays. The runtime needs no graphics processor, no WebAssembly and no framework, so it never competes with the game's renderer; it runs in the simulation's own worker thread, answers each distinct question once, and spreads a burst of questions over successive simulation steps rather than stalling any one of them. Around the forward pass the same principle holds: each snapshot is validated and keyed once

rather than for every question, and the option tokens' first-layer work is memoized, which together more than halve the student's total time when it replays the 18,040 questions of a forty-year town, to roughly 70 microseconds a question with every overhead included.

The same structure limits training cost. Full teacher distributions supply more information per label than a hard choice, on-policy collection supplies states reached under the student's decisions, and ordinary updates can warm-start from published weights. The isolated strike extension instead froze existing parameters so its effect on prior kinds could be checked exactly.

Set against the alternatives, these decisions account for the results of Section 6.1. The factorization machine is faster still, at about 17 microseconds, because it computes only fixed pairwise products, but it cannot let a third fact change the strength of a pair, and it pays for that in accuracy. Gradient-boosted trees come close in accuracy but share nothing across kinds and need forty times the space. The Deep Sets model shares the savings that come from the input: each fact's encoding and a town's pooled sum can be kept just as the set model keeps its keys and values, so once a town's facts have been seen it answers in about 80 microseconds, against about 70; but it runs its per-fact network on every new fact, which makes it about four times slower with nothing cached, and its smaller configurations trade accuracy for speed. The set model is the most accurate student, at a cost that is a small fraction of the game's own simulation.

Table 5. Where the evaluated core student's parameters go, and which of them each question uses. Times are per question on one processor core of the test machine.

Part	Parameters	Share	Work per question
Value embeddings	89,856	74.0%	one row looked up per fact
Scope, attribute, number and option embeddings	16,352	13.5%	rows looked up
Decision layer and scores	**12,993**	**10.7%**	**about 70 µs with the town's keys and values cached, or its facts met before; about 200 µs with nothing cached**
Error estimate	2,209	1.8%	a small network on the answer tokens' final states

7.3 The three-token layout

The Triple World Model's original layout, which writes each triple as three separate tokens, learns more slowly here and is set aside. In its original setting a state holds several entities sharing the same attributes, and relating values to entities across tokens is the problem attention solves. In Hearthvale each scope describes a single entity, so summing a triple's three embeddings already binds them, and the three-token form must instead learn that binding from position, at three times the cost per question. The representation transfers; the tokenization does not need to.

7.4 Evaluating a policy that acts repeatedly

The evaluation design deserves emphasis. A game samples the student's distribution repeatedly, so a small persistent answer bias can change a town more than larger random errors. We therefore compare answer probabilities and complete-town outcomes, including within-town spread, in

addition to divergence. The year-150 spread result shows why a nonsignificant mean difference is insufficient. Player behavior is simulated rather than observed, and the current long-run evidence does not cover strike resolution or the distinct live hosted `d1` policy. Future evaluation should directly measure those paths.

7.5 The teacher's character

The experiments reveal something about the teacher itself. Its answers keep towns markedly smaller than the fallback rules do, chiefly because it sends few residents to work. The student copies this faithfully, which is the intended behaviour, but it is a property of the teacher that the game's designers may wish to examine, and one that a distilled student makes cheap to study.

7.6 Limitations

The complete-town comparison uses four towns and six runs per policy; it cannot establish equivalence or exclude small mean effects. The two-run architecture comparison is less precise. The simulated player is only an approximation of human play, headless towns synthesize resident needs, and questions from play so far cover few players. The withheld-pair test covers six selected pairs rather than the full space of interactions. Timings come from one fast machine. The error estimator misses more than half of the worst questions at a 5% cutoff. Strike-specific evidence uses 397 held-out questions from 72 observed snapshots; no strike occurred in the short canary towns, and neither long-run strike outcomes nor strike confidence calibration have been measured. The distinct hosted `d1` service has not been shown to match the d1-3B reference teacher.

8 Conclusion

A compact set model of structured triples can carry most of Hearthvale's repeated decisions in the browser. On its evaluated core kinds it closely matches a d1-3B reference teacher's answer distributions, handles the withheld fact pairs tested and yields complete-town means for which the study found no difference, while showing wider year-150 population spread. The released bundle adds strike resolution with held-out answer-level evidence and exact replay parity on existing kinds; its long-run effect remains to be measured. A daily, gated learning pipeline supplies a way to improve coverage while keeping each play session on one bundle. More broadly, the study supports structured policy distillation where decision inputs and option sets are constrained, with whole-simulation testing needed to assess accumulated behavior.

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Appendix A. Detailed results

Table A1. Divergence on held-out towns and on withheld pairs, from a training run with the six pairs removed: absolute error on the withheld rows, and its ratio to the same model's error on ordinary held-out questions.

Student	Held-out towns	Withheld pairs	Ratio
Set model	**0.0238**	**0.0248**	**1.04×**
Set model, small	0.0288	0.0332	1.15×
Deep Sets	0.0271	0.0268	0.99×
Factorization machine, third order	0.0324	0.0366	1.13×
Factorization machine	0.0372	0.0427	1.15×
Pooled MLP (ablation)	0.0583	0.2194	3.76×
Gradient-boosted trees (five kinds)	—	0.0223	—

Table A2. Divergence on questions from teacher-governed towns, for students trained on the first stage of data only and on both stages, overall and weighted by how often each question is asked.

Student	Held-out towns	First stage: teacher-governed	First stage: by traffic	Both stages: teacher-governed	Both stages: by traffic
Set model, two layers per stage	0.0253	0.0232	0.0145	—	—
Set model, one layer per stage	**0.0258**	**0.0242**	**0.0164**	**0.0204**	**0.0122**
Set model, small	0.0294	0.0292	0.0184	0.0265	0.0155
Factorization machine	0.0314	0.0321	0.0185	0.0265	0.0156
Factorization machine, third order	0.0303	0.0311	0.0193	—	—
Pooled MLP (ablation)	0.0620	0.0613	0.0336	—	—

Table A3. Complete towns: the student's and the fallback rules' runs against the teacher's runs of the same four towns, six of each, played by the simulated player for 150 years. Shift is the mean difference from the teacher, with a 90% bootstrap interval and a permutation p-value; spread is the within-town standard deviation over the teacher's, with a 90% interval. Population is compared on a logarithmic scale.

Outcome	Student shift	p	Student spread	Fallback shift	p	Fallback spread
Student and fallback rules against the teacher						
Population, year 50	+0.08 [−0.17, +0.34]	0.678	0.75 [0.49, 1.06]	+1.17 [+0.97, +1.39]	<0.001	0.36 [0.26, 0.50]
Population, year 100	+0.28 [−0.04, +0.59]	0.170	0.94 [0.66, 1.39]	+1.68 [+1.40, +1.95]	<0.001	0.76 [0.37, 1.24]
Population, year 150	+0.07 [−0.21, +0.34]	0.709	1.54 [1.12, 2.30]	+1.52 [+1.23, +1.79]	<0.001	1.56 [0.74, 2.50]
Morning work share (points)	+0.32 [−0.60, +1.25]	0.628	1.08 [0.77, 1.61]	+35.78 [+35.02, +36.53]	<0.001	0.54 [0.37, 0.80]
Evening work share (points)	+0.00 [−0.18, +0.19]	0.976	0.75 [0.54, 1.13]	+5.71 [+5.54, +5.88]	<0.001	0.66 [0.46, 1.05]
Wellbeing, year 150	+2.50 [−1.33, +6.71]	0.322	0.78 [0.23, 2.30]	+3.04 [−1.71, +7.92]	0.348	1.13 [0.49, 3.27]

Table A4. Questions marked by the error estimate, on teacher-governed towns.

Questions marked	Share of all error	Share of the worst 5%	Random marking
top 1%	9.9%	15.6%	1%
top 5%	**26.7%**	**46.2%**	**5%**
top 10%	43.0%	69.2%	10%

Table A5. Training and deployment settings.

Setting	Value
Set model (evaluated core)	width 32, no snapshot encoder, one decision layer, two attention heads, feed-forward width 64; 119,201 parameters and an error estimate of 2,209
Released strike extension	22 supported kinds; shared core and confidence frozen; 288 strike-specific parameters trained; 397 held-out strike questions
Optimiser	AdamW, one-cycle schedule, peak learning rate 0.003, 20 epochs, batch 1,024; best epoch on held-out towns
Question weight	square root of how often simulated towns asked the question
Held-out towns	20% of towns, chosen by a hash of the town's seed
Bundle weights	8-bit integers with one scale per row; error estimate in 32-bit floats
Browser budget	4 ms of student work per turn of the event loop
Live referrals	eligible paid sessions only; per-account and global server quotas; local student or rules remain available when limited
Loop intake	one report suffices; at most 400 questions per client per cycle, the rest deferred; unusual states at most a quarter of a cycle unless three clients report them; questions already trained on skipped; a daily scheduled cycle with a minimum intake of 50 eligible questions
Loop gates	no more than 3% worse on teacher-governed towns overall or by traffic; no frequent kind more than 20% worse; largest answer bias no more than 5 points, overall and in hamlets and villages; rare-answer ratios between 0.67 and 1.5

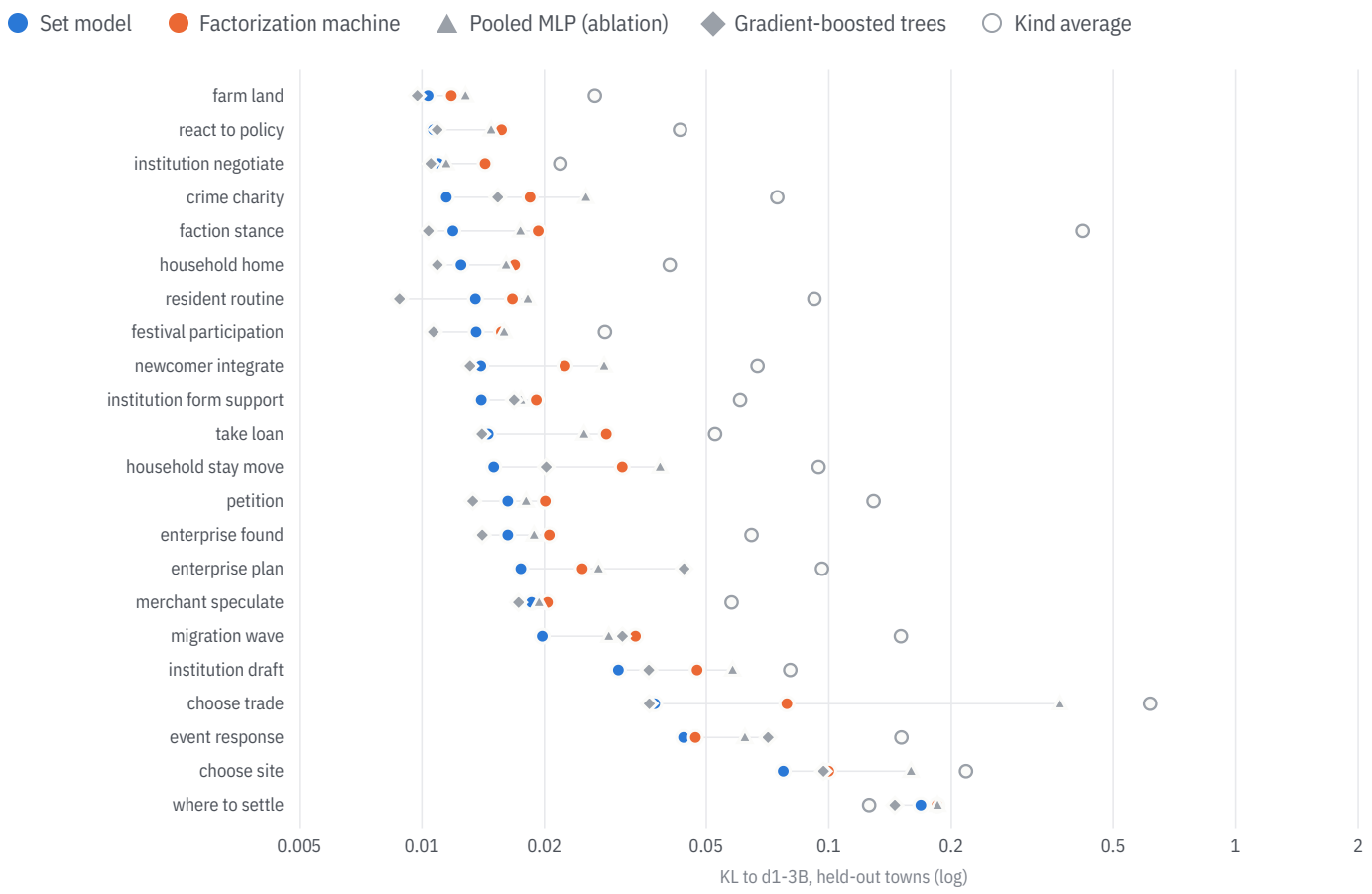


Figure A1. Divergence by kind of decision on held-out towns of the first stage (logarithmic scale). The hardest kinds are those whose options are free text that changes from question to question.

Reproducibility: decision features and browser inference are in `src/decision-features.js`, `src/decision-student.js` and `src/decision-local.js`; the Cloudflare training workflow and vendored training code are in `cloudflare/training/`. The released strike benchmark is `benchmarks/strike-resolve-v7.json`. Exported JavaScript fixtures check each bundle against its training implementation; the strike bundle's largest recorded difference was 2.53×10^{-7} on 64 fixtures.